Cross-Modal Transfer of the Risset Rhythm: Inducing Temporal Auditory Illusions in the Somatosensory System

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***Abstract*— The Risset rhythm is a well-known auditory illusion that creates the perception of a continuously accelerating tempo through overlapping rhythmic loops. While auditory illusions have been extensively studied, their translation to other sensory modalities remains largely unexplored. This study investigates whether the Risset rhythm can be successfully transferred from the auditory to the haptic domain using linear resonant actuator (LRA) motors. We conducted a pilot study with 6 participants who experienced five different vibration patterns and rated their perception of tempo change. Results show a significant effect of vibration pattern on perceived tempo acceleration ($p = 0.004$), and post-hoc analysis showed that the Risset rhythm was faster than a slowly accelerating ($p < 0.001$) and a constant vibration rhythm ($p < 0.001$). These preliminary findings suggest successful cross-modal transfer of the Risset rhythm to the somatosensory system.**

I. Introduction

Perceptual illusions are powerful tools for understanding how humans process sensory data. The Risset rhythm, created by Jean-Claude Risset [1], produces an illusion of a continuously accelerating tempo despite a constant underlying periodicity, achieved through overlapping rhythmic patterns that fade in and out systematically. This rhythm is primarily used in psychoacoustic exploration and in experimental musical compositions. While the Risset rhythm has been extensively studied in the auditory domain, its translation to the tactile domain remains largely unexplored.

Advances in haptic technology, such as linear resonant actuators (LRAs) and synthesis platforms like Syntacts [2], enables precise control over vibration stimuli with temporal characteristics due to LRA's higher, more consistent, and faster-starting vibration frequencies. Previous work has demonstrated the integration of haptic and auditory rhythms [3], and that temporal patterns can even be communicated through vibrotactile stimulation [4].

This study investigates if the auditory Risset rhythm illusion can be transferred to the sense of touch. We developed haptic equivalents of the Risset rhythm and varying tempo change profiles, and assessed participants' perceived acceleration under haptic-only and audio-haptic conditions.

II. Methods

A. Experimental Setup

The experimental setup utilized Syntacts boards to convert audio waveforms into haptic signals [2]. LRAs (Digikey Motor VG0832012) delivered the vibrotactile stimuli to participants' fingertips (Fig. 1). A custom user interface was developed in Python using the Tkinter library to present the audio stimuli and collect the participant ratings.

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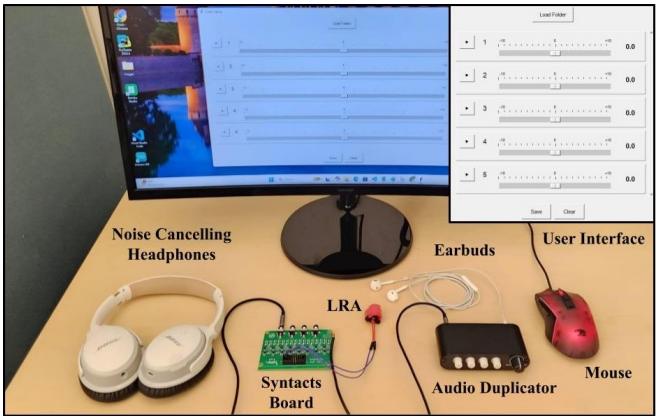

Fig. 1. Experimental setup showing Syntacts board, LRA motor powered by a Syntacts Board, and the study UI.

Five different temporal patterns were created, each lasting 8 seconds: (1) constant 60 bpm throughout (Constant), (2) acceleration from 60 to 80 bpm (Slow), (3) acceleration from 60 to 90 bpm (Medium), (4) acceleration from 60 to 100 bpm (Fast), and (5) the Risset rhythm illusion (constant underlying tempo with illusory acceleration, Risset). The Risset rhythm was adapted from an online source [5], and the first 8 seconds of the audio were cropped to create the experiment stimulus. The haptic Risset stimulus was a direct vibrotactile rendering of the original audio waveform. This raw audio waveform was played directly through LRA through the Syntacts board without any additional filtering, frequency modification, envelope shaping, or other signal processing. The first pattern served as a control condition with no actual tempo change, while patterns 2-4 provided stimuli with real acceleration. The Risset rhythm was the test condition for illusory perception.

B. User study

A pilot study was conducted with 6 participants (ages 18-25; three females, three males to evaluate the effectiveness of risset rhythm in haptics. The protocol was approved by the UCLA Institutional Review Board (25-2197), and all participants gave consent. Each participant completed three experimental trials in the same order: (1) haptics (H), (2) audio with haptics (AH), and (3) a second set of haptics only (H2). Because the illusion's audio effectiveness had already been confirmed, the AH condition was introduced after the initial haptics trial and combined with haptics to

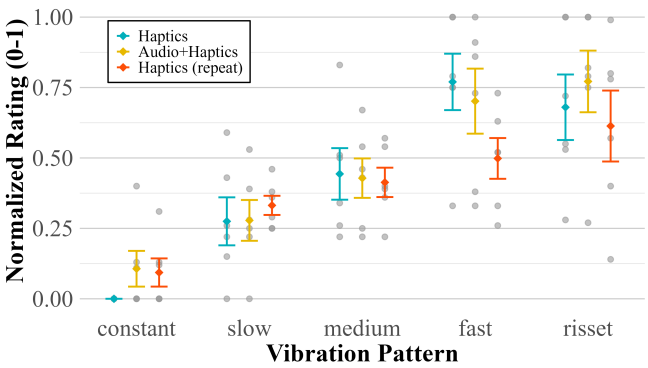

Fig. 2. Individual normalized ratings by vibration pattern and trial. Mean and standard error are shown in blue (Haptics, H), yellow (Audio with Haptics, AH), and red (Haptics set 2, H2). Shades of gray reflect individual participant data. The three trial conditions are shown for each of the five vibration patterns.

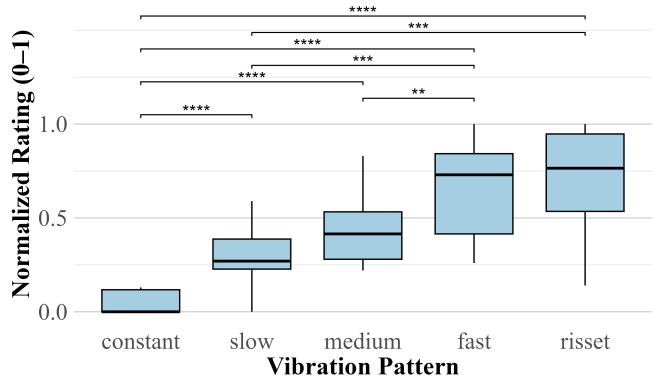

Fig. 3. Normalized perception ratings by vibration pattern. Box plots show the median, interquartile range, and whiskers. Standard significance notation is used for p-values (**: $0.001 < p < 0.01$; ***: $0.0001 < p < 0.001$; ****: $p < 0.0001$). Ratings systematically increased from constant (60-60 bpm) to fast (60-100 bpm) patterns, with the Risset rhythm rated similarly to the fast pattern.

assess enhanced learning via audio reinforcement. Within each trial, participants experienced all five vibration patterns in a randomized order and rated each vibration on their perceived tempo change from -10 (strongly decreasing tempo) to 0 (no tempo change) to +10 (strongly increasing tempo). To prevent perception of any LRA sound artifacts, noise-cancelling headphones were worn during haptics-only conditions (H, H2). During AH conditions, participants used earbuds to receive auditory stimuli.

C. Data Analysis

The study yielded 90 total observations (6 participants × 3 trials × 5 vibration patterns). Due to highly subjective ratings, min-max normalization was used to account for individual differences in scale usage: $x_{\text{norm}} = (x - x_{\min})/(x_{\max} - x_{\min})$. This procedure transformed each participant's data to a common 0-1 range while preserving relative differences across conditions. Across all participants, raw ratings ranged from 0 to 8.1, with individual response spans varying between 3 to 8 scale units. A 3 x 5 repeated measures ANOVA was conducted with trial condition (H, AH, H2) and vibration pattern (Constant, Slow, Medium, Fast, Risset). Post-hoc pairwise comparisons were performed to examine the differences between vibration patterns. Statistical analysis and plotting were performed in RStudio (R 4.5.2).

III. Results

Main Effects: Figure 2 shows the mean and standard error of the acceleration perception ratings across all conditions. The repeated-measures ANOVA revealed a significant main effect of vibration pattern ($F(1.47, 7.36) = 14.11$, $p = .004$, $\eta^2_G = 0.738$), indicating that different temporal patterns elicited different perceptions of tempo acceleration. As shown in Figure 3, post-hoc comparisons showed that perceived acceleration increased with actual tempo change, with the constant (60-60) control condition rated lowest and the fast (60-100) condition rated highest. More importantly, the Risset rhythm pattern was perceived as accelerating despite maintaining a constant periodicity, demonstrating the successful transfer of the auditory illusion to haptics.

In contrast, there was no significant main effect of trial condition ($F(2, 10) = 0.93$, $p = .43$, $\eta^2_G = 0.157$).Although this pattern suggests no measurable difference between haptic-only and audio-haptic trial conditions, as shown in Figure 2, the fixed order (H → AH → H2) limits the interpretation of this effect. Since participants experienced these conditions in this specific sequence, potential learning effects may confound the modality differences. These current findings cannot conclusively determine whether adding audio enhanes tempo change perception.

Interaction Effects: The interaction between trial and pattern was not significant, ($F(8, 40) = 1.94$, $p = .08$, $\eta^2_G = 0.279$), though there was a trend suggesting that audio may affect tempo perception differently depending on the specific rhythmic pattern. Further investigation with larger sample sizes may be meaningful.

IV. Discussion and Future Work

This study shows evidence for successful cross-modal transfer of the Risset rhythm from the auditory to the haptic domain. Participants reliably perceived the illusory tempo acceleration in haptic Risset rhythms while distinguishing between different degrees of genuine tempo change, validating the experimental model. The haptic implementation of the Risset rhythm opens new paths for future research. Other temporal and spatial illusions (e.g., Shepard tone) could be tested for cross-model transfer. Future studies will include larger and more diverse participant pools.

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